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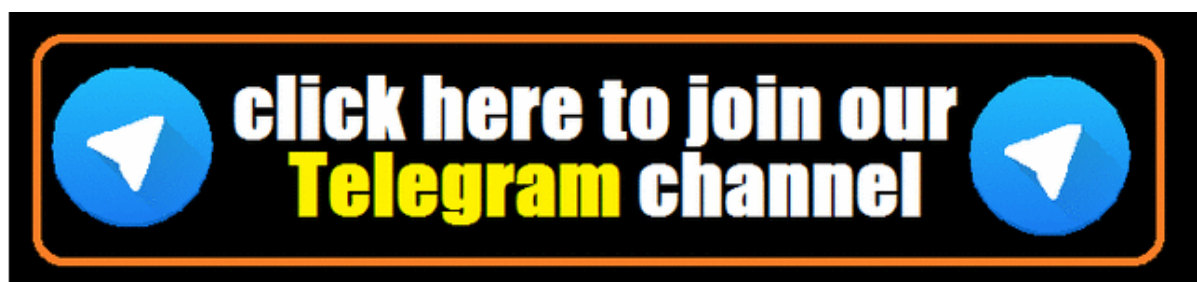
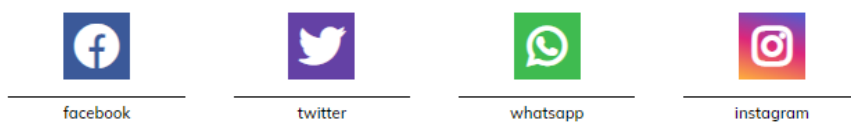


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## IMPORTANT FACTS AND FORMULAE FOR JEE

### AIEEE – PHYSICS

#### UNITS & MEASUREMENT

1. If  $x = a^m b^n c^p$ ,  
then fractional error in 'x' can be calculated as
$$\frac{\Delta x}{x} = m \frac{\Delta a}{a} + n \frac{\Delta b}{b} + p \frac{\Delta c}{c}.$$
2. For vernier callipers, least count =  $s - v$ .  
( $s$  = length of one division on main scale,  $v$  = length of one division on vernier scale.)
3. Length measured by vernier calliper = reading of main scale + reading of vernier scale  $\times$  least count.
4. For Screw Gauge, least count =  $\frac{\text{pitch of the screw}}{\text{no. of divisions on the circular scale}}.$   
where, Pitch =  $\frac{\text{distance travelled on the pitch scale}}{\text{no. of rotations}}.$
5. Length measured by screw gauge = Reading of main scale + Reading of circular scale  $\times$  least count.

#### MOTION IN ONE DIMENSION

- |          |   |                           |       |   |                |
|----------|---|---------------------------|-------|---|----------------|
| $t_1$    | = | initial time,             | $t_2$ | = | final time     |
| $u$      | = | initial velocity,         | $v$   | = | final velocity |
| $v_{av}$ | = | average velocity          | $a$   | = | acceleration   |
| $s$      | = | net displacement/distance |       |   |                |
1. Average speed,  $v_{av} = \frac{\text{total distance travelled}}{t_2 - t_1}$
  2. Instantaneous speed,  $v = \left| \lim_{\Delta t \rightarrow 0} \left( \frac{\Delta s}{\Delta t} \right) \right| = \left| \frac{ds}{dt} \right|$

$$3. \quad \text{Displacement} = \int_{t_1}^{t_2} \vec{v} dt$$

$$4. \quad \text{Total distance} = \int_{t_1}^{t_2} |v| dt$$

$$5. \quad a = \frac{dv}{dt} = v \frac{dv}{ds} = \frac{d^2s}{dt^2}$$

6. When acceleration is constant,

$$V = u + at, \quad s = ut + \frac{1}{2}at^2 \quad \& \quad v^2 = u^2 + 2as$$

### MOTION IN TWO AND THREE DIMENSION

$\omega$  = angular velocity,  $\alpha$  = angular acceleration,

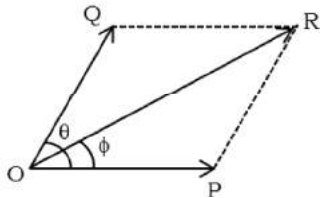
$a_t$  = tangential acceleration,  $a_n$  = normal acceleration

$R$  = radius of curvature

### Vectors

$$1. \quad \left| \vec{R} \right| = \sqrt{P^2 + Q^2 + 2PQ \cos \theta}$$

Where ' $\theta$ ' is angle between the vectors and direction of  $\vec{R}$  from  $\vec{P}$ ,



$$\phi = \tan^{-1} \frac{Q \sin \theta}{P + Q \cos \theta}$$

2. Two vectors  $\vec{a} \left( a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k} \right)$  and  $\vec{b} \left( b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k} \right)$  are equal if :

$$a_1 = b_1,$$

$$a_2 = b_2 \text{ and}$$

$$a_3 = b_3$$

3. If angle between two vectors  $\vec{a}$  and  $\vec{b}$  is ' $\theta$ '

$$\vec{a} \cdot \vec{b} = ab \cos \theta \text{ and}$$

$$\vec{a} \times \vec{b} = (ab \sin \theta \hat{n}, (\hat{n} \text{ is unit vector perpendicular to both } \hat{a} \text{ and } \hat{b}))$$

4. Velocity of 'B' with respect to 'A',

$$\vec{V}_{BA} = \vec{V}_B - \vec{V}_A$$

### Projectile Motion

5. Time of flight,  $t_0 = \frac{2u \sin \theta}{g}$

6. Range,  $R = \frac{u^2 \sin 2\theta}{g}$

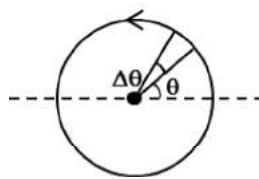
7. Maximum height,  $H = \frac{u^2 \sin^2 \theta}{2g}$

8.  $(x, y) = \left( u \cos \theta t, u \sin \theta t - \frac{1}{2} g t^2 \right)$

9. Equation of projectile,  $y = x \tan \theta - \frac{g x^2}{2u^2 \cos^2 \theta}$

### Circular Motion

10.  $\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta \theta}{\Delta t} = \frac{d\theta}{dt}$



$$\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$$

$$11. \quad a_{\tau} = \frac{dv}{dt}$$

$$a_N = \frac{v^2}{R}$$

$$a_{\text{Total}} = \sqrt{a_{\tau}^2 + a_N^2}$$

$$12. \quad R \text{ at } (x, y) = \frac{\left(1 + \left(\frac{dy}{dx}\right)^2\right)^{3/2}}{\frac{d^2y}{dx^2}}$$

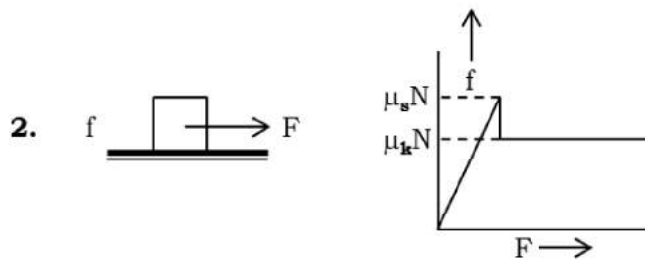
13. Banking of roads,

$$\tan \theta = \frac{v^2}{rg}$$

$$14. \quad \text{Centripetal force} = \frac{mv^2}{r} = m\omega^2 r$$

#### **LAWS OF MOTION**

$$1. \quad \begin{array}{c} \leftarrow R \quad \boxed{\bullet} \quad \rightarrow F \end{array} \quad \bullet \rightarrow a \quad F - R = ma$$



where  $f$  is the frictional force.

$$3. \quad \frac{\Delta p}{\Delta t} = F$$

$$4. \quad \vec{P} = \sum m_i \vec{V}_i$$

## WORK, ENERGY AND POWER

1. Total energy = kinetic energy + potential energy

2. Kinetic Energy =  $\frac{1}{2}mv^2$ , Potential Energy =  $mgh$

3. 
$$\int dW = \int_i^f \vec{F} \cdot d\vec{r}$$

4. Conservative forces : spring force, electrostatic force, gravitation force

Non-conservative forces : frictional force, viscous force

5. For elastic collision.

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

$$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$$

6. For perfectly inelastic collision,

$$V = \frac{m_1u_1 + m_2u_2}{m_1 + m_2}$$

7. If coefficient of restitution is  $e$  ( $0 < e < 1$ ),

Velocity of separation =  $e$  (velocity of approach)

8. Impulse =  $\int_{t_1}^{t_2} \vec{F} dt = \vec{P}_2 - \vec{P}_1$

## ROTATIONAL MOTION AND MOMENT OF INERTIA

$\tau$  = torque,  $F$  = force

$I$  = moment of inertia  $\alpha$  = angular acceleration

$L$  = angular momentum  $a_{cm}$  = acceleration of centre of mass

1. 
$$\vec{R}_{cm} = \frac{\sum_i m_i \vec{r}_i}{M}$$

2. 
$$r_{cm} = \frac{1}{M} \int r dm$$

3.  $\vec{\tau} = \vec{r} \times \vec{F}$

4. Moment of inertia,

$$I = \sum_i m_i r_i^2 = \int r^2 dm$$

$$\vec{\tau}_{\text{ext}} = I \vec{\alpha}$$

5. Angular momentum,

$$\vec{L} = \vec{r} \times \vec{p}$$

$$\vec{L} = I \vec{\omega}$$

6.  $\frac{d\vec{L}}{dt} = \vec{\tau}_{\text{ext}}$

7. (i) Pure translation  $\rightarrow \vec{\tau}_{\text{CM}} = 0$

(ii) Rotation  $\rightarrow \vec{\tau}_{\text{CM}} \neq 0$

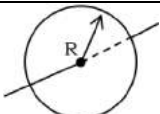
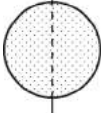
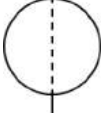
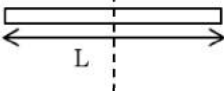
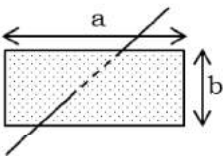
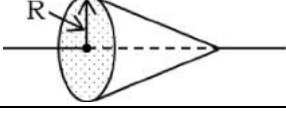
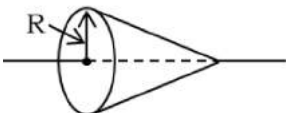
(iii) Pure rotation  $\rightarrow \sum \vec{F} = 0, \vec{\tau}_{\text{CM}} \neq 0$

(iv) Translation  $\rightarrow \sum \vec{F} \neq 0$

(v) Rolling  $\rightarrow a_{\text{CM}} = \alpha r, V_{\text{CM}} = \omega r$

(vi) Sliding or sliding  $\rightarrow a_{\text{CM}} \neq \alpha r, V_{\text{CM}} \neq \omega r$

**Table for standard moments of inertia :**

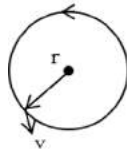
|    |                                                                                            |                                                                                      |                                 |
|----|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------|
| 1. | Very thin circular hoop                                                                    |    | $MR^2$                          |
| 2. | Uniform solid cylinder about its symmetry axis                                             |    | $\frac{1}{2}MR^2$               |
| 3. | Uniform solid sphere about its diameter                                                    |    | $\frac{2}{5}MR^2$               |
| 4. | Uniform hollow sphere about its diameter                                                   |    | $\frac{2}{3}MR^2$               |
| 5. | Uniform thin rod about its centre and perpendicular to it                                  |   | $\frac{M\ell^2}{12}$            |
| 6. | Thin rectangular sheet about an axis perpendicular to sheet and passing through its centre |  | $\frac{M(a^2 + b^2)}{12}$       |
| 7. | Uniform solid right cone about its axis                                                    |  | $\left(\frac{3}{10}\right)MR^2$ |
| 8. | Uniform hollow right cone about its axis                                                   |  | $\frac{1}{2}MR^2$               |

### GRAVITATION

G = Universal gravitational constant,

U = Gravitational potential energy

V = Gravitational potential,

- $E$  = Gravitational field  
 $F$  = Gravitational force
1.  $F = G \frac{Mm}{r^2}$  (attraction force)
  2. Gravitational potential energy,  $V = -G \frac{m_1 m_2}{r}$
  3.  $U_f - U_i = - \int_i^f \vec{F} \cdot d\vec{r}$
  4. Gravitational potential,  $V = -\frac{GM}{r}$
  5. Gravitational field,  $E = \frac{F}{m} = \frac{GM}{r^2}$
  6. Escape velocity,  $u \geq \sqrt{\frac{2GM}{R}}$
  7. (i)  $g' = g \left( 1 - \frac{h}{R_e} \right)$ , where 'h' is depth from the earth's surface.  
 (ii)  $g' = \frac{g}{\left( 1 + \frac{h}{R_e} \right)^2}$ , where 'h' is height from the earth's surface.
  8.  $v = \sqrt{\frac{GM}{r}}$
  9.  $T = 2\pi \sqrt{\frac{r^3}{GM}}$  
  10. K.E. =  $\frac{GMm}{2a}$ , P.K.E. =  $-\frac{GMm}{a} \Rightarrow E = -\frac{GMm}{2a}$
  11. Gravitational field,  $E = \frac{F}{m} = \frac{GM}{r^2}$

(i) Uniform solid sphere

$$E(r) = \frac{GMr}{R^3}, \quad r < R$$

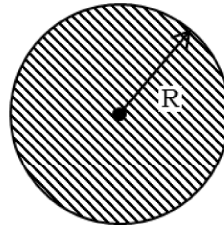
$$E(r) = \frac{GM}{R^2}, \quad r = R$$

$$E(r) = \frac{GM}{r^2}, \quad r > R$$

$$V(r) = -\frac{GM}{R} \left[ \frac{3}{2} - \frac{1}{2} \frac{r^2}{R^2} \right], \quad r < R$$

$$V(r) = -\frac{GM}{R}, \quad r = R$$

$$V(r) = -\frac{GM}{r}, \quad r > R$$



(ii) Uniform spherical shell

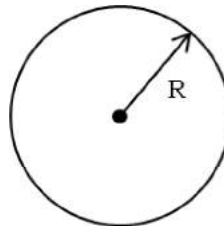
$$E(r) = 0, \quad r < R$$

$$E(r) = \frac{GM}{R^2}, \quad r = R$$

$$E(r) = \frac{GM}{r^2}, \quad r > R$$

$$V(r) = -\frac{GM}{R}, \quad r \leq R$$

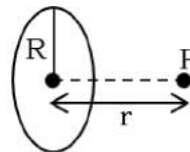
$$V(r) = -\frac{GM}{r}, \quad r > R$$



(iii) Uniform circular ring at point on axis :

$$E(r) = \frac{GMr}{(R^2 + r^2)^{3/2}}$$

$$V(r) = -\frac{2GM}{R^2} \left[ \sqrt{R^2 + r^2} - R \right]$$



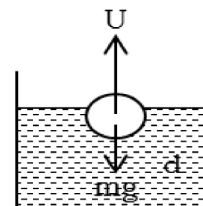
## SOLIDS AND FLUIDS

|                                    |                                 |
|------------------------------------|---------------------------------|
| $p$ = pressure,                    | $\rho$ = density                |
| $V$ = volume of solids,            | $v$ = volume immersed           |
| $D$ = density of solid,            | $d$ = density of liquid         |
| $A$ = cross section area           | $U$ = upthrust                  |
| $Y$ = Young's modulus,             | $\sigma$ = stress               |
| $\varepsilon$ = strain             | $B$ = Bulk modulus              |
| $F$ = force                        | $A$ = cross section area        |
| $\ell$ = initial length            | $\Delta\ell$ = change in length |
| $T$ = surface tension,             | $R$ = radius of the bubble/drop |
| $r$ = radius of the tube           | $\theta$ = angle of contact     |
| $\eta$ = coefficient of viscosity, | $F$ = force                     |

1.  $p = \lim_{\Delta S \rightarrow c} \frac{F}{\Delta S}$

2. Variation of pressure with height,  $dP = -\rho g dh$

3. Archimedes Principle,  $mg = v dg$  or  $VD = dv$



4. Equation of continuity,  $A_1v_1 = A_2v_2$

5. Bernoulli's equation,  $P + \frac{1}{2}\rho v^2 + \rho gh$

6.  $Y = \frac{\sigma}{\varepsilon} = \frac{F\ell}{A\Delta\ell}$

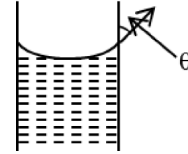
7. Bulk modulus,  $B = -\frac{\Delta P}{\Delta v/V}$

8. Elastic potential energy =  $\frac{1}{2} \times \text{stress} \times \text{strain} \times \text{volume}$

9. Excess pressure inside a drop,  $\Delta P = \frac{2T}{R}$

10. Excess pressure inside a soap bubble,  $\Delta P = \frac{4T}{R}$

11. Rise of liquid in a capillary tube,  $h = \frac{2T \cos \theta}{r \rho g}$



12.  $F = -\eta A \frac{dv}{dz}$

13. Stoke's law,  $F = 6\pi\eta r v$

14. Terminal velocity,  $v_0 = \frac{2r^2(\rho - \sigma)g}{9\eta}$

where,  $v$  = velocity,  $\sigma$  = density of liquid,  $\rho$  = density of solid

15. Velocity of efflux =  $V = \sqrt{2gh}$

### OSCILLATIONS

$\omega$  = angular frequency,  $I$  = moment of inertia

$T$  = time period,  $\ell$  = length of pendulum

1.  $\frac{d^2x}{dt^2} + \omega^2 x = 0$

2.  $T = \frac{2\pi}{\omega}$

3. Simple pendulum,

$$T = 2\pi\sqrt{\ell/g}, \quad \omega = \sqrt{g/\ell}$$

4. Angular simple harmonic motion,

$$T = 2\pi\sqrt{I/k}, \quad \omega = \sqrt{\frac{k}{I}}$$

5. Physical pendulum

$$T = 2\pi\sqrt{\frac{I}{mg\ell}}, \quad \omega = \sqrt{\frac{mg\ell}{I}}$$

**6. Effective spring constant**

i) Parallel  $K_{\text{eff}} = K_1 + K_2$

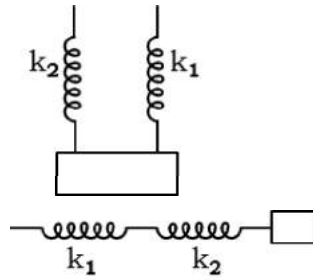
**7. Effective spring constant**

i) Parallel

$$k_{\text{eff}} = k_1 + k_2$$

ii) Series

$$\frac{1}{k_{\text{eff}}} = \frac{1}{k_1} + \frac{1}{k_2}$$



**WAVES**

|              |                      |            |                        |
|--------------|----------------------|------------|------------------------|
| A            | = amplitude          | $\mu$      | = mass per unit length |
| y            | = displacement       | $\omega$   | = angular frequency    |
| $\Delta\phi$ | = phase difference   | $\Delta x$ | = path difference      |
| v            | = frequency          | $\lambda$  | = wavelength           |
| L            | = length of the wire |            |                        |

**1. Equation of wave,**

$$y = A \sin \omega \left( t - \frac{x}{v} \right)$$

**2. Velocity of a wave on a string,**

$$v = \sqrt{\frac{F}{\mu}} \quad (\mu = \text{mass per unit length})$$

**3.  $P_{\text{av}} = 2\pi^2 \mu v A^2 \gamma^2$**

**4.  $y_1 = a_1 \sin \omega t$ ,  $y_2 = a_2 \sin(\omega t + \phi)$**

resultant wave,  $y = y_1 + y_2$

consecutive interference,  $\Delta\phi = 2n\pi$  or  $\Delta x = n\lambda$

destructive interference,  $\Delta\phi = (2n-1)\pi$  or  $\Delta x = \left(n - \frac{1}{2}\right)\lambda$

5.  $v_0 = \frac{1}{2L} \sqrt{\frac{F}{\mu}}$ , ( $v_0$  = fundamental frequency)
6. Speed of sound in fluid =  $\sqrt{\frac{B}{\rho}}$  (B = Bulk modules  $\rho$  = density)
7. Speed of sound in solid =  $\sqrt{\frac{Y}{\rho}}$  (Y = Young's modules  $\rho$  = density)
8. Speed of sound in gas =  $\sqrt{\frac{\gamma P}{\rho}}$ ,  $v \propto \sqrt{T}$
9. Closed organ pipe,  $v = (2n + 1) \frac{v}{4\ell}$
10. Open organ pipe,  $v = \frac{nv}{\ell}$
11. Frequency of beats =  $|v_1 - v_2|$
12. Doppler effect,  $v = \frac{v + u_o}{v - u_s} v_0$

#### HEAT AND THERMODYNAMICS

|                                                |                                            |
|------------------------------------------------|--------------------------------------------|
| P = pressure                                   | V = volume                                 |
| n = no. of moles                               | T = temperature                            |
| R = universal gas constant                     | $\alpha$ = coefficient of linear expansion |
| $\beta$ = coefficient of superficial expansion | $\gamma$ = coefficient of volume expansion |
| Q = heat taken/supplied                        | s = specific heat                          |
| m = mass                                       | $\Delta\theta$ = change in temperature     |
| L = latent heat of state change per mass       | W = work done by gas                       |
| U = internal energy                            | $V_i$ = initial volume                     |
| $V_f$ = final volume                           | $P_i$ = initial pressure                   |
| $P_f$ = final pressure                         |                                            |

1. Ideal gas equation,  $PV = n R T$
2. Thermal expansion,  $\alpha = \frac{1}{L} \frac{dL}{dT}$ ;  $\gamma = \frac{1}{V} \frac{dV}{dT}$
3.  $v_{\text{rms}} = \sqrt{\frac{3PV}{M}} = \sqrt{\frac{3RT}{M_0}}$   
 Translational kinetic energy,  
 $k = \frac{3}{2} PV$   
 $\bar{v} = \sqrt{\frac{8RT}{\pi M_0}}$  ;  $v_p = \sqrt{\frac{2RT}{M_0}}$
4. Vander Waals equation :  

$$\left( P + \frac{n^2 a}{V^2} \right) (V - nb) = nRT$$
5.  $Q = ms \Delta \theta$
6. Latent heat of state change,  
 $Q = mL$
7.  $dQ = dW + dU$
8. Work done by a gas,  

$$W = \int_1^2 P dV$$
9. Work done on an ideal gas :  
 isothermal process,  $W = nRT \ln \left( \frac{V_f}{V_i} \right)$   
 isobaric process,  $W = P(V_f - V_i)$   
 isochoric process,  $W = 0$   
 adiabatic process,  $W = \frac{P_i V_i - P_f V_f}{\gamma - 1}$

10. Entropy,

$$\Delta S = \frac{\Delta Q}{T}$$

11. Efficiency of an ideal reversible heat engine,

$$\eta = 1 - \frac{T_2}{T_1}$$

12.  $C_v = \left( \frac{\Delta Q}{n\Delta T} \right)$  constant volume

$$C_p = \left( \frac{\Delta Q}{n\Delta T} \right) \text{ constant pressure}$$

13.  $dU = nC_v dT$

14.  $C_p - C_v = R, \frac{C_p}{C_v} = \gamma$

15. Adiabatic process,  $PV^\gamma = \text{constant}$

#### TRANSFERENCE OF HEAT

|   |   |                                     |          |   |                   |
|---|---|-------------------------------------|----------|---|-------------------|
| e | = | Coefficient of emission             | $\sigma$ | = | Stefan's constant |
| K | = | Coefficient of thermal conductivity | A        | = | Area              |

1.  $\frac{\Delta Q}{\Delta t} = K \frac{A(T_1 - T_2)}{x} = -KA \frac{dT}{dx}$

2. Thermal resistance,  $R = \frac{x}{KA}$

Heat current,  $I = \frac{T_1 - T_2}{R}$

3. Series connection,  $R = R_1 + R_2$

Parallel connection,  $1/R = 1/R_1 + 1/R_2$

4.  $U = e \sigma A T^4$ , (U = energy emitted per second)

5. Newton's law of cooling,

$$\frac{d\theta}{dt} = -K(\theta - \theta_0) \quad (\theta \text{ taken in Celsius scale})$$

## ELECTROSTATICS

### Facts :

1. Charge is quantized.
2. Charge is conserved, net charge does not change.
3. The tangent to a line of force at any point gives the direction of electric field at that point.
4. No electric field exists inside conductors.
5. Electric fields at surfaces of conductors are perpendicular to surface.
6. All points on or inside a conductors are at the same potential.
7. In isolated capacitor, charge does not change.
8. Capacitors in series have equal amount of charges.
9. The voltage across two capacitor connected in parallel is same.
10. In steady state no current flows through a capacitor.

|           |   |                             |        |   |                       |
|-----------|---|-----------------------------|--------|---|-----------------------|
| F         | = | force                       | q      | = | charge                |
| $\lambda$ | = | linear charge density       | C      | = | capacitance           |
| V         | = | electric potential          | E      | = | electric field        |
| $\sigma$  | = | surface charge density      | $\rho$ | = | volume charge density |
| U         | = | electrical potential energy | r      | = | distance              |
| $q_{enc}$ | = | charge enclosed             | p      | = | dipole moment         |

1. Coulombs law : The force between two point charges at rest,

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}$$

2. Electric field :

$$\vec{F} = q \vec{E}$$

3. Electric Potential :

$$\Delta V = -E \Delta r \cos\theta$$

$$V = - \int_{\infty}^{\vec{r}} \vec{E} \cdot d\vec{r}$$

4. Electric Potential Energy :

$$U = \frac{1}{4\pi\epsilon_0} \sum_{i < j} \frac{q_i q_j}{r_{ij}}$$

5. Fields for a point charge :

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2},$$

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r},$$

Gauss's law : net electric flux through any closed surface is equal to the net charge enclosed by the surface divided by  $\epsilon_0$ .

$$\oint_s \vec{E} \cdot d\vec{s} = \frac{q_{enc}}{\epsilon_0}$$

Electric Dipole : It is a combination of equal and opposite charges.

Electric dipole moment  $\vec{P} = q \vec{d}$ , where  $d$  is the separation between the two point charges.

**Electric field due to various charge distribution :**

a) If the line charge is ' $\lambda$ ' then

i) electric field outside cylinder (Radius ' $R$ ')

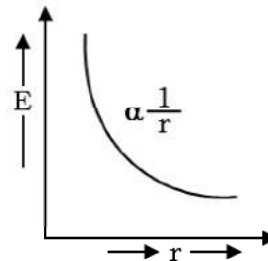
$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

ii) electric field on the surface

$$E = \frac{\lambda}{2\pi\epsilon_0 R}$$

iii) Electric field inside at a distance ' $r$ ' from the axis

$$E = \frac{\lambda r}{2\pi\epsilon_0 R^2}$$



b) Plane sheet of charge :

$$E = \frac{\sigma}{2\epsilon_0}, \text{ where } \sigma \text{ is the surface charge density.}$$

c) Near a charge conducting surface :

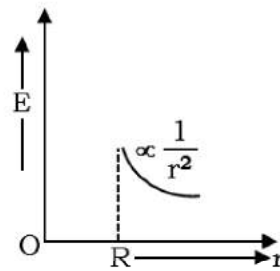
$$E = \frac{\sigma}{\epsilon_0}$$

d) Charged conducting spherical shell :

$$\vec{E}_{\text{in}} = 0, r < R$$

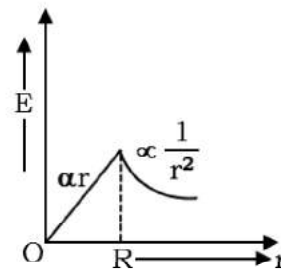
$$\vec{E}_{\text{out}} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}, r > R$$

$$\vec{E}_{\text{sur}} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}, r = R$$



e) Non conducting charged solid sphere :

$$E_{\text{in}} = \frac{1}{4\pi\epsilon_0} \frac{qr}{R^3}, r < R$$



$$\vec{E}_{\text{out}} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}, r > R$$

$$\vec{E}_{\text{sur}} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}, r = R$$

f) Charged ring :

Special cases :-

$$E_{\text{axial}} = \frac{1}{4\pi\epsilon_0} \frac{qx}{(R^2 + x^2)^{3/2}}$$

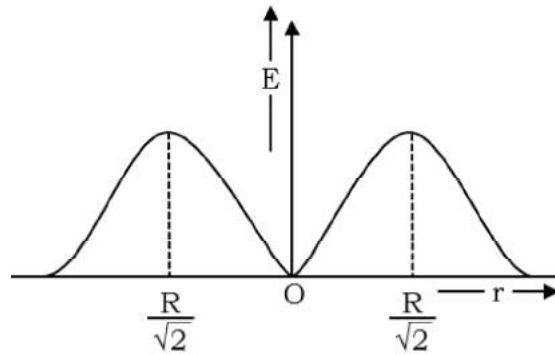
i) If  $x < R$  and  $x^2 \ll R^2$ ,

$$E = \frac{1}{4\pi\epsilon_0} \frac{qx}{R^3},$$

ii) If  $x > R$  and  $x^2 \gg R^2$ ,

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{x^2}$$

Note :  $E_{\text{max}}$  at  $r = R/\sqrt{2}$ .



g) Dipole :

In terms of polar coordinates,

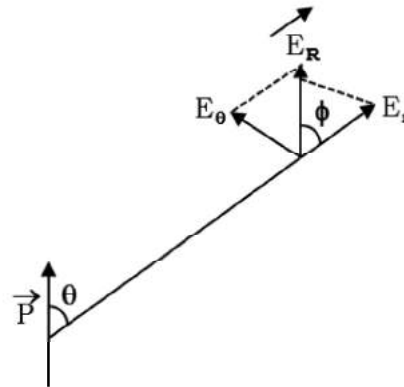
$$E_r = \frac{1}{4\pi\epsilon_0} \frac{2P \cos \theta}{r^3}$$

$$E_\theta = \frac{1}{4\pi\epsilon_0} \frac{P \sin \theta}{r^3}$$

$$E_R = \sqrt{E_r^2 + E_\theta^2}$$

$$E_R = \frac{P}{4\pi\epsilon_0 r^3} \sqrt{1 + 3 \cos^2 \theta}$$

Its direction,  $\phi = \tan^{-1} \left( \frac{\tan \theta}{2} \right)$



**Torque on a dipole :**

$$\vec{\tau} = \vec{P} \times \vec{E}$$

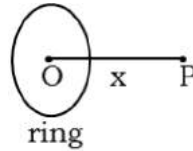
**Potential energy of Dipole :**

$$U = -\vec{P} \cdot \vec{E}$$

**Electric Potential due to various charge distribution :**

a) Charged ring :

i) At the centre:-  $V = \frac{1}{4\pi\epsilon} \frac{q}{R}$

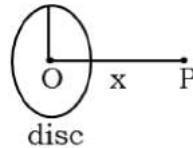


ii) On the axis at a distance x from the centre:-  $V = \frac{1}{4\pi\epsilon} \frac{q}{\sqrt{R^2 + x^2}}$

b) Uniformly charged disc :

i) At a distance x from the centre,

$$V = \frac{\sigma}{2\epsilon} \left[ \sqrt{R^2 + x^2} - x \right]$$



ii) At the centre,

$$V = \frac{\sigma R}{2\epsilon}$$

iii) At a large distance x ( $\gg R$ ),

$$V = \frac{1}{4\pi\epsilon} \frac{q}{x}$$

Here, the disc behaves as a point charge.

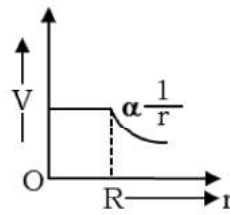
iv) At the edge of the disc,

$$V = \frac{\sigma R}{\epsilon}$$

c) Spherical shell :

$$V_{\text{in}} = \frac{1}{4\pi\epsilon} \frac{q}{R}$$

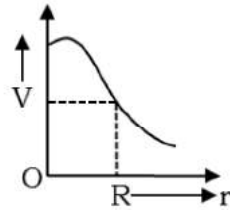
$$V_{\text{out}} = \frac{1}{4\pi\epsilon} \frac{q}{r}$$



d) Nonconducting sphere :

$$V_{\text{in}} = \frac{1}{4\pi\epsilon} \frac{q}{R} \left[ \frac{3}{2} - \frac{r^2}{2R^2} \right]$$

$$V_{\text{sur}} = \frac{1}{4\pi\epsilon} \frac{q}{R}; V_{\text{out}} = \frac{1}{4\pi\epsilon} \frac{q}{r}$$



6. Capacitance of a parallel plate capacitor,

$$C = \frac{\epsilon_0 A}{d}$$

7. Voltage across capacitor,

$$V = \frac{Q}{C}$$

8. Elements in series :

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}, R = R_1 + R_2$$

9. Elements in parallel :

$$C = C_1 + C_2; \frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

10. If a dielectric is placed inside the capacitor,

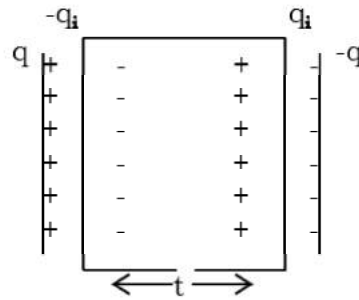
$$E = \frac{E_0}{K}, V = \frac{V_0}{K}$$

11. Induced charged in the dielectric,

$$q_i = q \left( 1 - \frac{1}{K} \right)$$

- 12.** Capacitance of a capacitor partially filled with dielectric of thickness  $t$ ,

$$C = \frac{\epsilon A}{\left(d - t + \frac{t}{K}\right)}$$

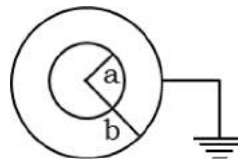


- 13.** Force between the plates of a capacitor

$$F = \frac{Q^2}{2\epsilon A}$$

- 14.** Capacitance of a spherical capacitor,

$$C = \frac{4\pi\epsilon ab}{(b-a)}$$

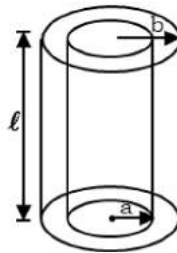


- 15.** Capacitance of a parallel plate capacitor,

$$C = \frac{\epsilon_0 A}{d}$$

- 16.** Capacitance of a cylindrical capacitor,

$$C \text{ (per unit length)} = \frac{2\pi\epsilon}{\ln\left(\frac{b}{a}\right)}$$



### Numbers :

Electronic charge,  $e = 1.6 \times 10^{-19}$

$$\frac{1}{4\pi\epsilon} = 9 \times 10^9 \text{ Nm}^2 / \text{C}^2$$

$$\epsilon = 8.85 \times 10^{-12} \text{ C}^2 / \text{Nm}^2$$

### Units :

| Quantity       | Units    | Conversion                   |
|----------------|----------|------------------------------|
| Electric field | N/C, V/m | 1 N/C = 1 V/m                |
| Charge         | C        | 1e = $1.6 \times 10^{-19}$ C |
| Voltage        | V, J/C   | 1V = 1 J/C                   |

### CURRENT ELECTRICITY

#### Facts:

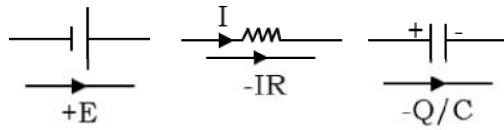
1. Sum of currents into a node is zero.
2. Sum of voltage around closed loop is zero.
3. The temperature coefficient of resistivity is negative for semiconductor.

|                      |                         |
|----------------------|-------------------------|
| Q = charge           | $v_d$ = drift velocity  |
| i, I = current       | j = current density     |
| $\rho$ = resistivity | R = resistance          |
| t = thickness        | K = dielectric constant |

#### Formulae :

1. Electric current,  
 $I = n q A v_d$
2. Resistance of a wire,  
 $R = \rho \frac{L}{A}$
3. Current density,  
 $J = \frac{I}{A} = n q v_d$
4. Voltage across the resistor,  
 $V = IR$
5. Ohm's law,  
 $E = \rho J$
6. Power dissipated in resistor :  
 $P = I^2 R = \frac{V^2}{R}$
7. Charge on a capacitor in an RC – circuit,  
 $Q(t) = Q_0 (1 - e^{-t/RC})$  where  $Q_0$  is the charge at  $t = 0$

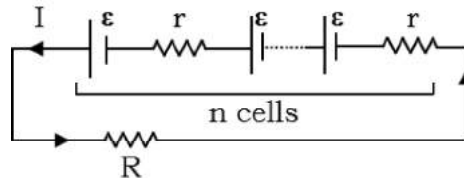
8. Current through a resistor in an RC – circuit,  
 $I(t) = I e^{-t/\tau}$ , where  $\tau = RC$



9. Grouping of cells :

- a) Series combination :-

$$I = \frac{n\varepsilon}{nr + R}$$

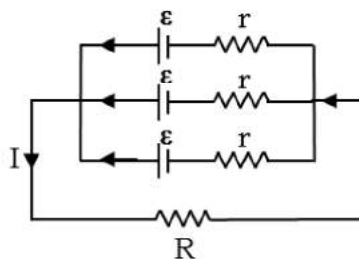


If the polarity of 'm' cells is reversed.

$$I = \frac{(n - 2m)\varepsilon}{(R + nr)}$$

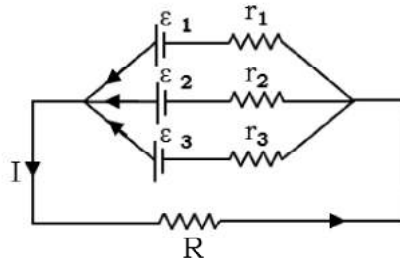
- b) Parallel combination :-

i)  $I = \frac{\varepsilon}{(R + r/n)}$



ii)

$$I = \frac{\sum \frac{E}{r}}{1 + R \sum \left(\frac{1}{r}\right)}$$

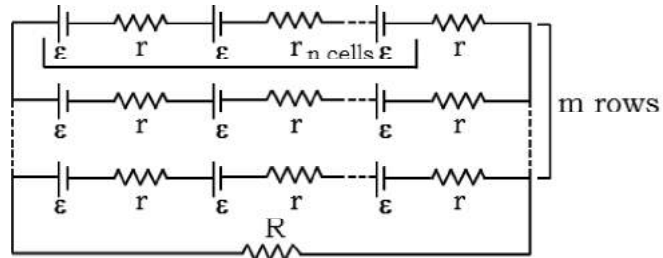


c) Mixed combination :-

$$I = \frac{nE}{\left(R + \frac{nr}{m}\right)}$$

Current will be maximum

when  $R = \frac{nr}{m}$

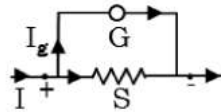


**Ammeter :**

$$S = \left( \frac{I_g}{I - I_g} \right) G$$

Percentage error in measuring current through an ammeter,

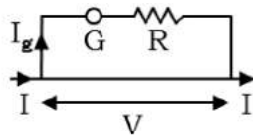
$$\% \text{Error} = \left( \frac{R'}{R + R'} \right) \times 100 \text{ where } R' = \frac{GS}{G + S}$$



**Voltmeter :**

$$R = \frac{V}{I_g} - G$$

$$\% \text{Error} = \left( \frac{1}{1 + \frac{r}{R + G}} \right) \times 100$$



### Potentiometer :

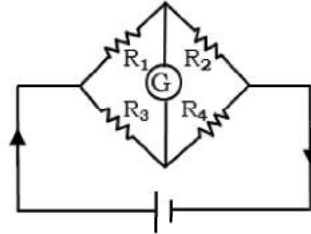
Comparing the emf of the cells,  $\frac{\epsilon_1}{\epsilon_2} = \frac{\ell_1}{\ell_2}$

Internal resistance of a battery,  $r = R \left( \frac{\ell}{\ell'} - 1 \right)$

### Wheatstone Bridge :

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

Under the above condition there will be no current through Galvanometer.



### Units :

| Quantity    | Units    | Conversion          |
|-------------|----------|---------------------|
| Capacitance | F        | 1 F = 1C/V          |
| Current     | A        | 1 A = 1C/S          |
| Resistance  | $\Omega$ | 1 $\Omega$ = 1V / A |

### THERMAL EFFECTS OF CURRENTS

#### Joules law of heating :

Heat produced in a resistor,

$$H = I^2 R t$$

#### Seeback Effect :

Thermo emf,  $E = \alpha t + \beta t^2$

### MAGNETIC EFFECTS OF CURRENTS

|                        |                                |
|------------------------|--------------------------------|
| B = magnetic induction | $\phi$ = flux                  |
| q = charge             | V = velocity                   |
| F = force              | N = no. of turns               |
| I = current            | $\mu$ = magnetic dipole moment |

**Facts :**

1. Force on a moving charge in magnetic field is  $\perp^r$  to both  $\vec{v}$  and  $\vec{B}$ .
2. Net magnetic force acting on any closed current loop in a uniform magnetic field is zero.
3. Magnetic field of long straight wire circles around wire.
4. Parallel wires carrying current in the same direction attract each other.

**Formulae :**

1. Force on a charged Particle,  $\vec{F} = q \left( \vec{v} \times \vec{B} \right)$   

$$F = qvB(\sin \theta)$$

2. When a particle enters into a  $\perp^r$  magnetic field, it describes a circle.  
 Radius of the circular path,

$$r = \frac{mv}{qB} = \frac{\sqrt{2Km}}{qB}$$

3. Time period,  $T = \frac{2\pi m}{qB}$

4. Magnetic force on a segment of wire,

$$\vec{F} = I \vec{\ell} \times \vec{B}$$

5. Force between parallel current carrying wires,

$$\frac{F}{\ell} = \frac{\mu}{2\pi} \frac{I_1 I_2}{r}$$

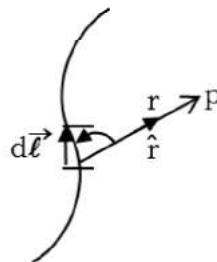
6. A current carrying loop behaves as a magnetic dipole of magnetic dipole moment,

$$\vec{\mu} = NIA \vec{\hat{A}}$$

7. Torque on a current loop,  $\vec{\tau} = \vec{\mu} \times \vec{B}$

8. Magnetic field due to a current carrying wire,

$$\vec{B} = \frac{\mu_0 I}{4\pi} \int \frac{d\vec{\ell} \times \vec{\hat{r}}}{r^2}$$



9. Ampere's law

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I_{\text{enc}}$$

**Magnetic Field due to various current distributions :**

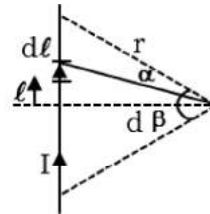
1) Current in a straight wire :-

$$B = \frac{\mu_0 I}{4\pi b} (\sin\alpha + \sin\beta)$$

2) For an infinitely long straight wire :-

$$\alpha = \beta = \pi / 2$$

$$B = \frac{\mu_0 I}{4\pi b}$$

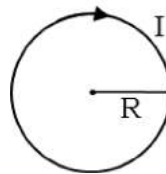


3) On the axis of a circular coil :-

$$B = \frac{\mu_0 N I R^2}{2(R^2 + x^2)^{3/2}}$$

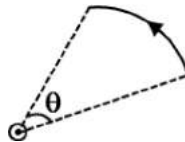
4) At the centre of the circular coil :-

$$B = \frac{\mu_0 I}{2R}$$



5) For a circular arc,

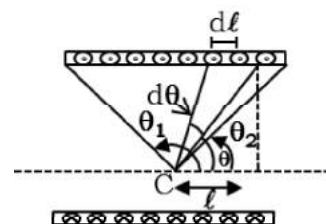
$$B = \frac{\mu_0 I}{4\pi R} \theta$$



6) Along the axis of a solenoid,

$$B_C = \frac{\mu_0 n I}{2} (\cos\theta_2 - \cos\theta_1)$$

where  $n = \frac{N}{\ell}$  (No. of turns per unit length)

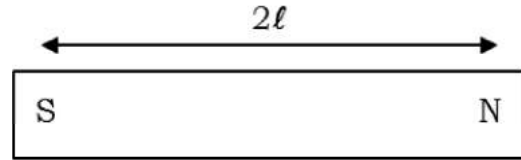


7) For a very long solenoid,  $B_c = \mu_0 nI$

8) At the end of a long solenoid,  $B = \frac{\mu_0 nI}{2}$

### MAGNETOSTATICS

1. Magnetic – Moment of a bar magnet :  
 $M = m \times 2\ell$  where 'm' is the pole Strength.



2. Force between two poles of strength  $m_1$  and  $m_2$  separated by a distance 'r' is given by,  $F = \frac{\mu_0}{4\pi} \frac{m_1 m_2}{r^2}$  where 'r' is in metre and m in Amp-metre.

3. Torque on a bar magnet placed in a magnetic field,

$$\vec{\tau} = \vec{M} \times \vec{B}$$

and the potential energy, 0

$$U = \vec{M} \cdot \vec{B}$$

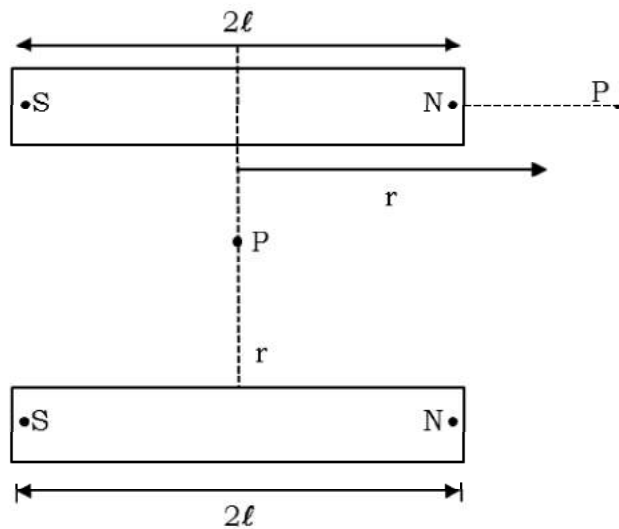
4. Magnetic field due to a bar magnet,

i)  $B_p = \frac{\mu_0}{4\pi} \frac{2M}{r^3}$

ii)  $B_p = \frac{\mu_0}{4\pi} \frac{M}{(r^2 + \ell^2)^{3/2}}$

If  $r \gg \ell$

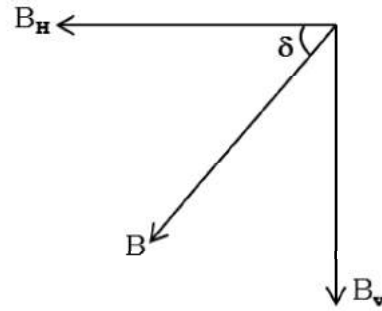
$$B = \frac{\mu_0}{4\pi} \frac{M}{r^3}$$



5. Earth's magnetic field,

$$\frac{B_H}{B} = \cos \delta$$

Where  $B_H$  is the horizontal Component of the earth's magnetic field and 'δ' is the angle of dip.



6. Magnetic intensity,

$$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{I}, \text{ where } I = \frac{M}{V}$$

7.  $\vec{B} = -\mu \vec{H}$

8.  $I = \chi \vec{H}$ , where  $\chi$  is the magnetic susceptibility.

9. Curie's law :  $\chi = \frac{C}{T}$

#### ELECTROMAGNETIC – INDUCTION AND ALTERNATING CURRENTS

|                                      |                                    |
|--------------------------------------|------------------------------------|
| B = magnetic induction               | $\phi$ = flux                      |
| q = charge                           | V = velocity                       |
| F = force                            | N = no. of turns                   |
| I = current                          | L = coefficient of self inductance |
| M = coefficient of mutual inductance | $\epsilon$ = induced emf           |

1. Induced emf,

$$\epsilon = -N \frac{d\Phi}{dt} = -NA \frac{dB}{dt} \text{ where } a \text{ is the area of a loop.}$$

2. Induced current,  $I = \frac{\epsilon}{R}$

3. Induced electric field,  $\oint \vec{E} \cdot d\vec{\ell} = -\frac{d\Phi}{dt}$

4. Self-inductance,

$$N\Phi = LI; \quad \epsilon = -L \frac{dI}{dt}$$

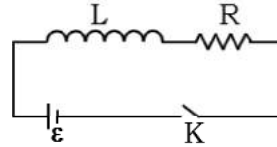
5. Inductance of a solenoid,  $L = \mu_0 n^2 A \ell$   
where 'n' is the number of turns per unit length.

6. Mutual inductance,  $N\Phi = MI$

$$\epsilon = -M \frac{dI}{dt}$$

7. Growth of current in an L-R circuit,

$$I = \frac{\epsilon}{R} (1 - e^{-t/\tau}), \tau = \frac{L}{R}$$



8. Decay of current in an L-R circuit,

$$I = \frac{\epsilon}{R} (e^{-t/\tau}), \tau = \frac{L}{R}$$

9. Energy stored in an inductor,  $U = \frac{1}{2} LI^2$

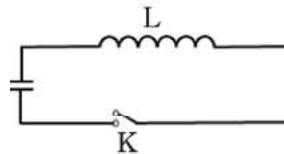
10. Energy density in a magnetic field,

$$u = \frac{U}{V} = \frac{B^2}{2\mu_0} \quad (\text{Energy per unit volume})$$

11. L-C-Oscillations :

$$\frac{d^2q}{dt^2} = -\left(\frac{1}{LC}\right)q$$

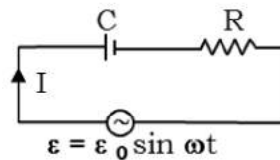
$$q = q_0 \cos \omega t \quad \text{where, } \omega = \frac{1}{\sqrt{LC}}$$



12. R.M.S. current,  $I_{\text{rms}} = \frac{I_0}{\sqrt{2}}$

13. In RC - circuit  
Peak current,

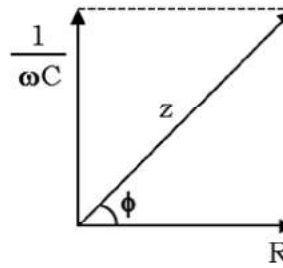
$$I_0 = \frac{\epsilon_0}{Z} = \frac{\epsilon_0}{\sqrt{R^2 + (1/\omega C)^2}} \quad 14.$$



14. Steady state current,

$$I = \frac{\epsilon_0}{z} \sin(\omega t + \phi)$$

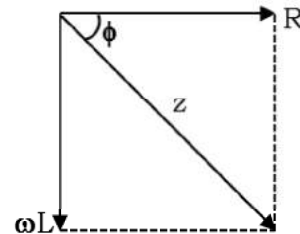
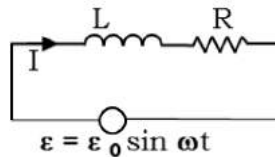
$$\phi = \tan^{-1} \left( \frac{1}{\omega CR} \right)$$



15. In LR – Circuit,  
In steady state,

$$I = \frac{\epsilon}{\sqrt{R^2 + \omega^2 L^2}} \sin(\omega t - \phi)$$

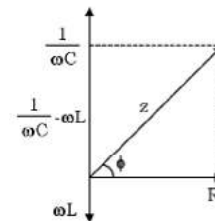
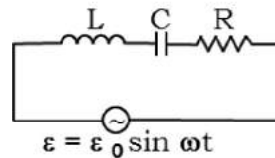
$$\phi = \tan^{-1} \left( \frac{\omega L}{R} \right)$$



16. In LCR – Circuit :

$$I = \frac{\epsilon}{\sqrt{R^2 + \left( \frac{1}{\omega C} - \omega L \right)^2}} \sin(\omega t - \phi)$$

$$\phi = \tan^{-1} \left( \frac{\frac{1}{\omega C} - \omega L}{R} \right)$$



- i) If  $\frac{1}{\omega C} > \omega L$ , current leads the voltage.
- ii) If  $\frac{1}{\omega C} < \omega L$ , current lags behind the voltage.
- iii) If  $\frac{1}{\omega C} = \omega L$ , current is in phase with the voltage.
- iv) In LCR – circuit, peak value of current,

$$I = \frac{\epsilon}{\sqrt{R^2 + \left( \frac{1}{\omega C} - \omega L \right)^2}}$$

If the angular frequency ' $\omega$ ' varies then  $I_0$  also varies.

$I_0$  is maximum, when  $\frac{1}{\omega C} - \omega L = 0$

$$\text{or } \omega = \frac{1}{\sqrt{LC}}$$

$$f = \frac{1}{2\pi\sqrt{LC}}$$

This is called the resonance frequency of the circuit.

**17. Power in an A. C. circuit,**

$P = V_{\text{rms}} I_{\text{rms}} \cos \phi$  where  $\cos \phi$  is the power factor.

i) For a purely resistive circuit,  $\phi = 0$

ii) For a purely reactive circuit,  $\phi = \pi/2$ , or  $-\pi/2$

$$\therefore \cos \phi = 0$$

**18. Transformer :**

$$\left| \frac{\epsilon_1}{\epsilon_2} \right| = \frac{N_1}{N_2} \text{ where } N_1 \text{ and } N_2 \text{ are the number of turns in the}$$

primary and secondary coils respectively.

**Electromagnetic - Induction**

$$\text{Induced electric field, } \oint \vec{E} \cdot d\vec{\ell} = -\frac{d\Phi}{dt}$$

$$\text{Motional emf} = \epsilon = \int_a^b (\vec{v} \times \vec{B}) \cdot d\vec{\ell}$$

**Modified Faraday's law**

$$\epsilon = \oint \underbrace{(\vec{v} \times \vec{B}) \cdot d\vec{\ell}}_{\substack{\text{Motion of a} \\ \text{conductor in a} \\ \text{stationary} \\ \text{magnetic field}}} = - \underbrace{\vec{A} \cdot \frac{d\vec{B}}{dt}}_{\substack{\text{Variation of} \\ \text{magnetic field} \\ \text{in a} \\ \text{stationary} \\ \text{conductor}}}$$

## RAY OPTICS

|                                                   |                               |
|---------------------------------------------------|-------------------------------|
| $u$ = distance of the object from the lens/mirror |                               |
| $v$ = distance of the image from the lens/mirror  |                               |
| $m$ = magnification                               | $i$ = angle of incident       |
| $r$ = angle of refraction/reflection              | $n$ = refractive index        |
| $\theta_c$ = critical angle                       | $\delta$ = angle of deviation |
| $R$ = radius of curvature                         | $P$ = power                   |

1. Spherical mirrors,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}, \quad m = -\frac{v}{u}$$

2. Refraction at plane surfaces,

$$n = \frac{\sin i}{\sin r} = \frac{\text{real depth}}{\text{apparent depth}}$$

$$\text{critical angle, } \theta_c = \sin^{-1}\left(\frac{1}{n}\right)$$

3. Refraction at spherical surface,

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}, \quad m = \frac{n_1 v}{n_2 u}$$

4. Refraction through thin lenses,

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}, \quad m = \frac{v}{u}$$

$$P = \frac{1}{f}, \quad (f \text{ should be in metre})$$

5. Prism,

$$r + r' = A,$$

$$\delta = i + i' - A$$

$$n = \frac{\sin \frac{A + \delta_m}{2}}{\sin \frac{A}{2}}$$

6. Refraction at plane surfaces, Normal shift =  $\left(1 - \frac{1}{n}\right)t$ , t is the thickness of the glass slab.
7. For small 'A' and 'T',  $\delta = (n - 1)A$
8. Combination of lens  $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} + \dots$
9. When lens are 'd' distance apart  $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$

### WAVE OPTICS

|                                              |                                |
|----------------------------------------------|--------------------------------|
| c = speed of light,                          | a = amplitude                  |
| $\eta$ = refractive index of the material,   | I = intensity                  |
| v = speed of light in the material,          | W = fringe width               |
| $\Delta x$ = path difference                 | $\lambda$ = wavelength         |
| $\delta$ = phase difference                  | d = distance between the slits |
| D = distance between the slit and the screen | t = thickness of glass slab    |

1.  $v = \frac{c}{\eta}$  where  
 c = speed of light,  
 $\eta$  = refractive index of the material and  
 v = speed of light in that material
2. Young's double slit experiment,  
 $a = \sqrt{a_1^2 + a_2^2 + 2a_1 a_2 \cos \delta}$ ,  $I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \delta$   
 for bright fringes,  $\Delta x = n\lambda$   
 for dark fringes,  $\Delta x = \left(n - \frac{1}{2}\right)\lambda$  }  $n = 1, 2, 3 \dots$   

$$\text{Fringes width} = \frac{D\lambda}{d}$$
3.  $v = \frac{c}{\mu}$ ,  $c = \frac{1}{\sqrt{\mu \epsilon}}$ ,  $v = \frac{1}{\sqrt{\mu \epsilon}}$

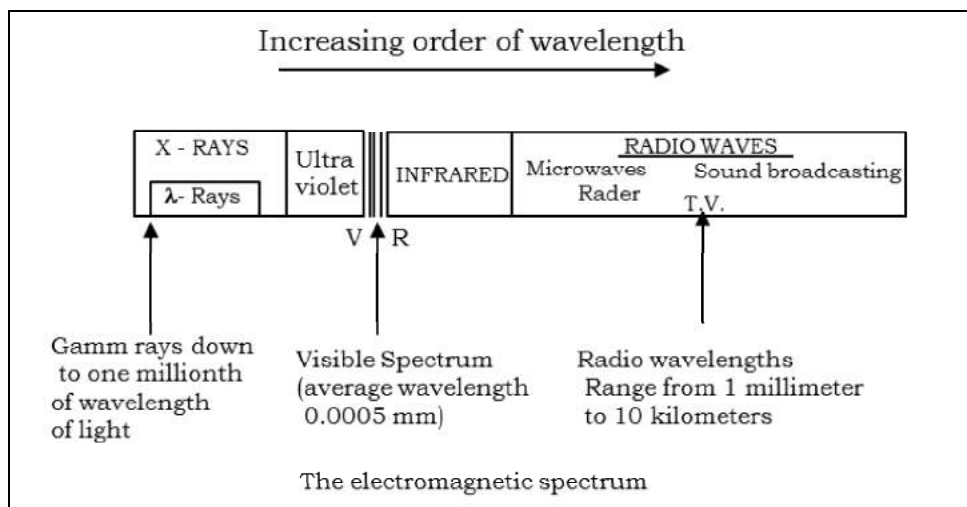
4. Young's double slit experiment,  
 $a = \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos \delta}$ ,  $I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \delta$   
 for bright fringes,  $\Delta x = n\lambda$   
 for dark fringes,  $\Delta x = \left(n - \frac{1}{2}\right)\lambda$   $n = 1, 2, 3, \dots$   

$$W = \frac{D\lambda}{d}$$
5. Shifting of fringes : Shift =  $\frac{(\mu - 1)tD}{d}$
6. Optical path length =  $\mu t$
7. Number of fringes shifted =  $\frac{\text{shift}}{\text{fringe width}} = \frac{(\mu - 1)t}{d}$
8. Dispersive Power,  $\omega = \frac{\mu_v - \mu_r}{\mu_y - 1}$
9. Dispersive without average deviation :  $\frac{A}{A'} = \frac{\mu'_y - 1}{\mu_y - 1}$
10. Average deviation without dispersion :  $\frac{A}{A'} = \frac{\mu'_v - \mu'_r}{\mu'_v - \mu'_r}$
11. Net average deviation  $\delta_r = (\mu_y - 1)A \left(1 - \frac{\omega}{\omega_r}\right)$

#### **ELECTROMAGNETIC WAVES**

1. Speed of electromagnetic waves,  $v = \frac{1}{\mu\epsilon}$
2. Electromagnetic waves and transverse,  $E = CB$

| Wave-Band             | Origin                                                                                                         | Sources                                                                                                                              |
|-----------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| X-radiation           | i) High energy changes in electron structures of atoms<br>ii) Decelerated electrons                            | X-ray tubes                                                                                                                          |
| Gamma radiation       | Energy changes in nuclei of atoms                                                                              | Radioactive substance                                                                                                                |
| Ultraviolet radiation | Fairly high energy changes in electron structure of atoms                                                      | i) Very hot bodies, e.g. the electric arc.<br>ii) Electric discharge through gases, particularly, mercury vapour in quartz envelopes |
| Visible Radiation     | Energy changes in electron structure of atoms                                                                  | Various lamps, flames and anything at or above red-heat                                                                              |
| Infrared Radiation    | Low energy changes in electron structure of atoms                                                              | All Matter over a wide range of temperature from absolute zero upwards                                                               |
| Radio wave            | i) High frequency oscillatory electric currents<br>ii) Very low energy changes in electron structure of atoms. | Radio transmitting circuits.                                                                                                         |



## ELECTRONS AND PHOTONS

|                        |                        |
|------------------------|------------------------|
| P = momentum           | h = Plank's constant   |
| $\lambda$ = wavelength | c = speed of light     |
| F = frequency          | $\phi$ = work function |

1. Momentum of a photon,  $P = \frac{h}{\lambda}$
2. Energy of a photon,  $E = hf = \frac{hc}{\lambda}$
3. Photoelectric equation,  $KE_{\max} = \frac{hc}{\lambda} - \phi$   
 $KE_{\max} = h(f - f_0)$
4. To take place photoelectric effect,  $\lambda \leq \lambda_0$
5. For a one electron atom,  $E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$   
 $r_n = 5.29 \times 10^{-11} \frac{n^2}{Z} \text{ meters}$
6. If an electron jumps from  $m^{\text{th}}$  orbit to  $n^{\text{th}}$  orbit,  
 $\frac{1}{\lambda} = RZ^2 \left[ \frac{1}{n^2} - \frac{1}{m^2} \right]$ , where  $m > n$
7. Mosley's law,  $\sqrt{f} = a(z - b)$  where a b are constants.
8. The minimum wavelength of x-rays,  $\lambda_{\min} = \frac{hc}{eV}$

## ATOMS, MOLECULES AND NUCEI

|                   |                       |
|-------------------|-----------------------|
| n = no. of nuclei | $t_{1/2}$ = half life |
| Z = atomic no.    |                       |

1. Nuclear radius,  $R = R_0 A^{1/3}$ ,  $R = 1.1 \times 10^{-15} \text{ metre}$
2. Mass energy equivalence,  $\Delta E = \Delta m C^2$
3. Radioactive decay,  $N(t) = N e^{-\lambda t}$  where  $\lambda$  is the decay constant.
4. Radioactive Half Life,  $t^{1/2} = \frac{0.693}{\lambda}$

### **SOLIDS AND SEMICONDUCTOR DEVICES**

1. Intrinsic semiconductors,  $n_i = n_e = n_h$
2. Doped semiconductors,  $n_i^2 = n_e n_h$
3. Conductivity,  $\sigma = e(n_e \mu_e + n_h \mu_h)$
4. Current gains,  $\alpha = \frac{\beta}{1 + \beta}, \beta = \frac{\alpha}{1 - \alpha}$
5. Alternating current power gain =  $\frac{\text{output power}}{\text{input power}}$